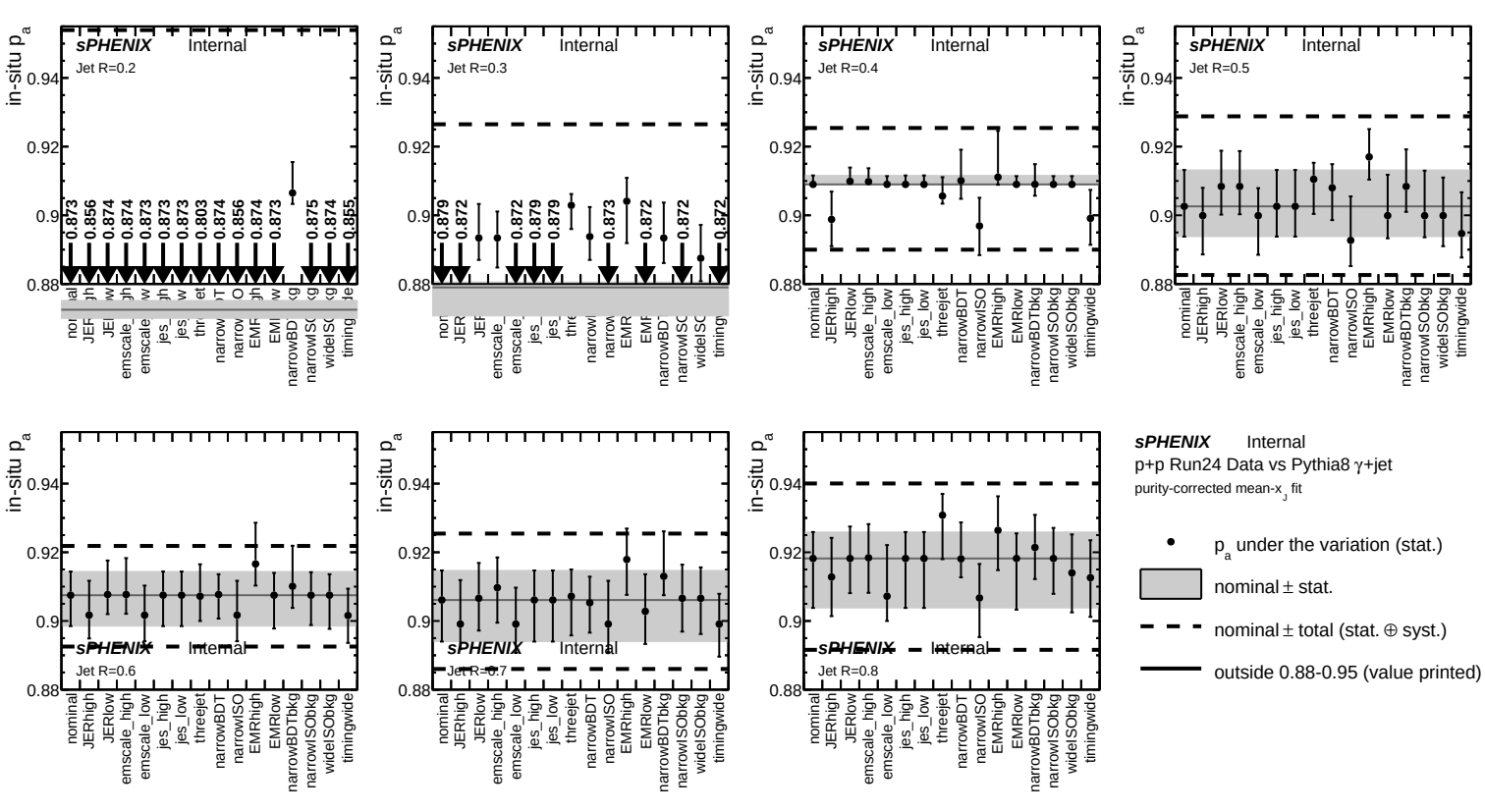
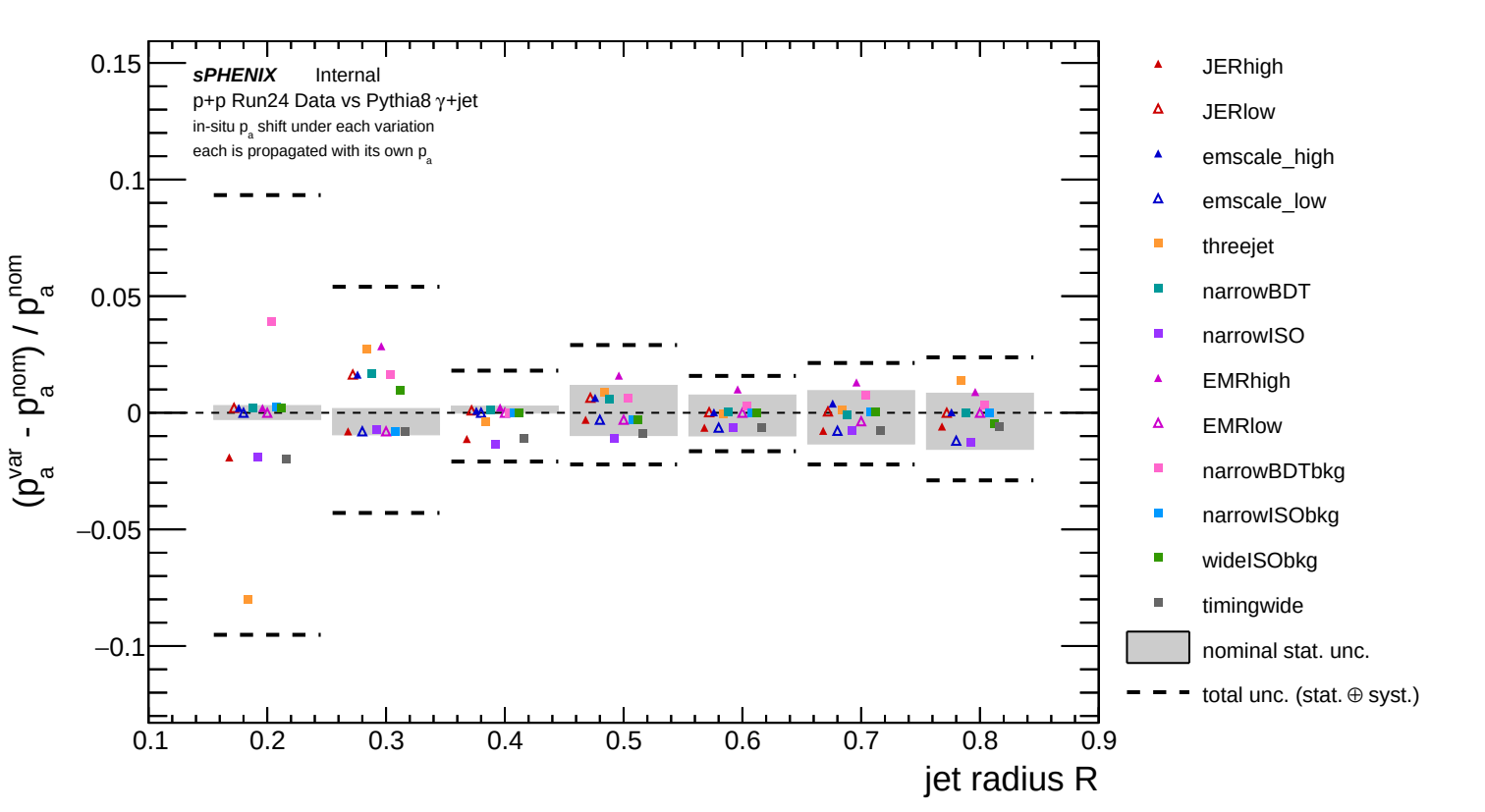

sPHENIX Internal
Jet R=0.2
Jet R=0.3
Jet R=0.4
Jet R=0.5
Jet R=0.6
Jet R=0.7
Jet R=0.8
in-situ p_a
0.94
0.92
0.9
0.88
0.873 0.856 0.874 0.874 0.873 0.873 0.873 0.803 0.874 0.856 0.874 0.873 0.874 0.875 0.874 0.865
0.879 0.872 0.879 0.879 0.872 0.873 0.872 0.872
nominal
JERhigh
JERlow
emscale_high
emscale_low
jes_high
jes_low
threejet
narrowBDT
narrowISO
EMRhigh
EMRlow
narrowBDTbkg
narrowISObkg
wideISObkg
timingwide
sPHENIX Internal
p+p Run24 Data vs Pythia8 γ+jet
purity-corrected mean-x_J fit
p_a under the variation (stat.)
nominal ± stat.
nominal ± total (stat. ⊕ syst.)
outside 0.88-0.95 (value printed)


sPHENIX Internal
p+p Run24 Data vs Pythia8 γ+jet
in-situ p_a shift under each variation
each is propagated with its own p_a
(p_a^var - p_a^nom) / p_a^nom
jet radius R
0.15
0.1
0.05
0
−0.05
−0.1
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
JERhigh
JERlow
emscale_high
emscale_low
threejet
narrowBDT
narrowISO
EMRhigh
EMRlow
narrowBDTbkg
narrowISObkg
wideISObkg
timingwide
nominal stat. unc.
total unc. (stat. ⊕ syst.)